METRIC

A-A-59441/6

16 January 2004

COMMERCIAL ITEM DESCRIPTION SPECIFICATION SHEET

VAPOR CORROSION INHIBITORS, TYPE VI

The General Services Administration has authorized the use of this commercial item description, for all federal agencies.

The complete requirements for procuring Type VI vapor corrosion inhibitors (VCIs) described herein shall consist of this document and the latest issue in effect of A-A-59441.

1. SCOPE. This commercial item description covers the requirements for Type VI VCIs to provide corrosion protection for metals used in mechanical, electrical and electronic components and equipment that are subjected to corrosive environments. Type VI VCIs are used in packaging applications for shipment and storage.

2. SALIENT CHARACTERISTICS.

2.1 Physical characteristics. See Table I.

TABLE I. Physical characteristics.

| **VAPOR CORROSION INHIBITORS -- TYPE VI** | | | | | |
|---|---|---|---|---|---|
| **Item** | **Description** | **Application** | **Protection Life (yr)** | **Protection Volume $m^3$ ($ft^3$)** | **Metals Protected** |
| 1 | Film, sheet, bag or tubing of transparent plastic, heat sealable VCI | Multipurpose packaging for corrosion protection of products, components or assemblies with metal surfaces | 5 | Volume enclosed by package, or 1$m^2$ (10 $ft^2$) metal area protected per 2-3 $m^2$ (20-30 $ft^2$) of total VCI area, or interior of 1$m^3$ (10 $ft^3$) of enclosed volume | Specify: F or NF or MM |
| 2 | Foam packaging sheet, open cell, VCI | Cushioning packaging for corrosion protection of heavy and irregularly shaped metal products, components or assemblies | 5 | Volume enclosed by package, or 1$m^2$ (10 $ft^2$) metal area protected per 2-3 $m^2$ (20-30 $ft^2$) of total VCI area, or interior of 1$m^3$ (10 $ft^3$) of enclosed volume | Specify: F or NF or MM |
| 3 | Foam packaging sheet, closed cell, dust and moisture resistant, VCI | Cushioning packaging for corrosion protection of heavy and irregularly shaped metal products, components or assemblies | 5 | Volume enclosed by package, or 1$m^2$ (10 $ft^2$) metal area protected per 2-3 $m^2$ (20-30 $ft^2$) of total VCI area, or interior of 1$m^3$ (10 $ft^3$) of enclosed volume | Specify: F or MM |



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TABLE I. Physical characteristics – Continued.

| Item | Description | Application | Protection Life (yr) | Protection Volume $m^3$ ($ft^3$) | Metals Protected |
|---|---|---|---|---|---|
| 4 | Bubble cushioning packaging sheet, VCI anticorrosion | Flexible cushion packaging with corrosion protection for fragile electronic, machined and other components or assemblies | 5 | Volume enclosed by package, or $1m^2$ (10 $ft^2$) metal area protected per 2-$3m^2$ (20-30 $ft^2$) of total VCI area, or interior of $1m^3$ (10 $ft^3$) of enclosed volume | Specify: F or MM |
| 5 | Multiple cross-ply high strength tear-resistant plastic film, bag, sheet, tarpaulin or tubing, heat sealable, VCI anticorrosion | Multipurpose packaging for corrosion protection of heavy and irregularly shaped metal products, components or assemblies | 5 | Volume enclosed by package, or $1m^2$ (10 $ft^2$) metal area protected per 2-$3m^2$ (20-30 $ft^2$) of total VCI area, or interior of $1m^3$ (10 $ft^3$) of enclosed volume | Specify: F or MM |
| 6 | Film, sheet, bag or tubing of transparent plastic, heat sealable, VCI anticorrosion and normal electrostatic dissipative (ESD) | Multipurpose packaging for corrosion protection and dissipation of electrostatic charge for electronics, munitions and other products, sub-assemblies and components | 1 (ESD), 5 VCI | Volume enclosed by package of VCI film or bag; surface resistivity nominal 1 x $10^{11}$, range $1x10^{10}$- $1x10^{12}\Omega$/sq ($1x10^{11}$-$1x10^{13}$ $\Omega$), and static decay < 0.1 sec to the 1% cutoff. | Specify: NF or MM |
| 7 | Film, sheet, bag or tubing of transparent plastic, heat sealable, VCI anticorrosion and permanent electrostatic dissipative (ESD) | Multipurpose long-life, packaging for corrosion protection and dissipation of electrostatic charge for electronics, munitions and other products, sub-assemblies and components | 5 | Volume enclosed by package of VCI film or bag; surface resistivity nominal 1 x $10^{11}$, range $1x10^{10}$- $1x10^{12}\Omega$/sq ($1x10^{11}$-$1x10^{13}$ $\Omega$), and static decay < 0.1 sec to the 1% cutoff. | Specify: NF or MM |
| 8 | Film, sheet, bag or tubing of transparent plastic, heat sealable, VCI anticorrosion and critical electrostatic dissipative (ESD) | Multipurpose packaging for corrosion protection and critical dissipation of electrostatic charge static-critical electronics components and sub-assemblies | 5 | Volume enclosed by package of VCI film or bag; surface resistivity nominal 1 x $10^{9}$, range $1x10^{8}$- $1x10^{10}\Omega$/sq ($1x10^{9}$-$1x10^{11}$ $\Omega$), and static decay < 0.01 sec to the 1% cutoff. | Specify: NF or MM |
| 9 | Paper, kraft, VCI saturated, #35, MIL-PRF-3420 | Protect ferrous metal products and components, as a wrap or interlayer | 5 | Volume enclosed by package, or $1m^2$ (10 $ft^2$) metal area protected per 2-$3m^2$ (20-30 $ft^2$) of total VCI area, or interior of $1m^3$ (10 $ft^3$) of enclosed volume | Specify: F or MM |



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TABLE I. Physical characteristics – Continued.

| Item | Description | Application | Protection Life (yr) | Protection Volume $m^3$ ($ft^3$) | Metals Protected |
|---|---|---|---|---|---|
| 10 | Kraft paper #35: base layer #30 paper, one side bonded to 1.5 mil polyethylene layer saturated with vapor and contact corrosion inhibitor, MIL-PRF-3420 | Packaging for corrosion protection of electrical/electronic and machines components, tools, and sub-assemblies | 5 | Volume enclosed by package, or 1$m^2$ (10 $ft^2$) metal area protected per 2-3 $m^2$ (20-30 $ft^2$) of total VCI area, or interior of 1$m^3$ (10 $ft^3$) of enclosed volume | Specify: F or MM |
| 11 | Paper, kraft, VCI coated, grease resistant, 35# | Protect ferrous metal products and components that require barrier to lubrication oils | 2 | Volume enclosed by package, or 1$m^2$ (10 $ft^2$) metal area protected per 2-3 $m^2$ (20-30 $ft^2$) of total VCI area, or interior of 1$m^3$ (10 $ft^3$) of enclosed volume | F |
| 12 | Waterproof, soft padded, UV resistant plastic sheet, bags or rolls, VCI anticorrosion | Packaging for high value marine and outdoor corrosion protection of products, components or assemblies with metal surfaces | 5 | Volume enclosed by package, or 1$m^2$ (10 $ft^2$) metal area protected per 2-3 $m^2$ (20-30 $ft^2$) of VCI paper area or interior of 1$m^3$ (10 $ft^3$) of enclosed volume | Specify: F or MM |
| 13 | Fabric, VCI coated, non-woven, LDPE | Protect high value products from corrosion and UV exposure during storage and shipment | 5 | Volume enclosed by package, or 1$m^2$ (10 $ft^2$) metal area protected per 1$m^2$ (10 $ft^2$) of VCI paper area or interior of 0.5 $m^3$ (5 $ft^3$) of enclosed volume | MM |

Key: MM = Multi-metal; F = Ferrous; NF = Non-ferrous

2.2 Application. Items one through three shall be used with conventional packaging materials. Items 4 through 10 shall incorporate the VCI protection into the packaging materials. Items nine and ten shall provide corrosion and limited physical protection for heavy equipment and aircraft, which must be shipped or stored while exposed to the environment.

2.3 Performance. VCI material shall not interfere with or modify the electrical, optical or mechanical surface properties of the materials within the packaging.



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3. National stock numbers (NSNs). Table II provides NSNs that correspond to this CID. The list may not be indicative of all possible NSNs associated with the CID.

TABLE II. National stock numbers.

| Item | Standard Package | NSN |
|---|---|---|
| 1 | Specify film or bag thickness 50-150µ (2 to 6 mil), roll width up to 6 m (20 ft), bag width up to 3 m (10 ft), whether bags are plain or gusseted, and whether made of normal (low density) polyethylene or HDPE (high density polyethylene) or other PE resin, and whether heat-shrink, non-shrink, or stretch properties. | TBD |
| 2 | Thickness 5 mm (120 mil), roll width 1m (3.1 ft), and sheets to 0.25 m square (0.78 ft). | TBD |
| 3 | Specify: Thickness 3 to 10 mm (30 to 250 mils); roll width .3 to 2 m (1 to 6 ft); and Length to 2000 ft. | TBD |
| 4 | Specify roll size up to width 1.22 m (4.0 ft) and bubble sizes from 3 to 13 mm (0.12 to 0.5 inches). | TBD |
| 5 | Specify film or bag thickness 80-500µ (3 to 12 mils), size up to width 4 m (12.5 ft), plain or gusseted bags and tubs, and whether heat-shrink, or non-shrink properties. | TBD |
| 6 | Specify film or bag thickness 50-150µ (2 to 6 mils), roll size up to width 6 m (20 ft), bag size up to width 3 m (10 ft), whether bags are plain or gusseted, and whether made of normal (low density) polyethylene or HDPE (high density polyethylene) or other PE resin, and whether heat-shrink, non-shrink, or stretch properties. | TBD |
| 7 | Specify film or bag thickness 50-150µ (2 to 6 mils), roll size up to width 6 m (20 ft), bag size up to width 3 m (10 ft), whether bags are plain or gusseted, and whether made of normal (low M8 density) polyethylene or HDPE (high density polyethylene) or other PE resin, and whether heat-shrink, non-shrink, or stretch properties. | TBD |
| 8 | Specify film or bag thickness 50-150µ (2 to 6 mils), roll size up to width 6 m (20 ft), bag size up to width 3 m (10 ft), whether bags are plain or gusseted, and whether made of normal (low density) polyethylene or HDPE (high density polyethylene) or other PE resin, and whether heat-shrink, non-shrink, or stretch properties. | TBD |
| 9 | Specify: roll or sheet width 0.91 m or 1.22 m (3 or 4 ft), length up to 3000 ft length, and non-standard weights 30# to 90#, PE coating 0.5 to 2.0 mil, and colors. | TBD |
| 10 | Specify: roll or sheet width 0.91 m or 1.22 m (3 or 4 ft), length up to 3000 ft length, and non-standard weights 30# to 90#, PE coating 0.5 to 2.0 mil, and colors. | TBD |
| 11 | Specify roll or sheet size either 0.91 m (3 ft) or 1.22 m (4 ft) width up to 200 ft length. | TBD |
| 12 | Specify thickness 100-500µ (4 to 20 mils), roll width up to width 1.9 m (6 ft). | TBD |
| 13 | Specify roll or sheet size, weight, e.g., 165 or 220 g/m$^2$, and closure method. | TBD |



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MILITARY INTERESTS

Custodians:
Navy – SH
Air Force – 99

Review Activities:
DLA – CC

CIVIL AGENCY COORDINATING ACTIVITY:
GSA - FSS

Preparing Activity:
Navy – SH
(Project 2090-0118-006)



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